

INSTRUMENTAL DRIFT Correction

Workflow Documentation

Relative gravimeters are inherently subject to instrumental drift, a phenomenon primarily driven by the mechanical relaxation and thermal sensitivity of the internal spring mechanism, and is typically less than 20 $\mu\text{Gal}/\text{hour}$ (Kennedy et al., 2021).

While software like **GSadjust** (used later for network adjustment) offers built-in drift correction functions, these routines typically require a survey design with frequent station repeats or closed loops. Since our campaign utilized a **star network** with a pivot station and **open loops** (see next section on network adjustment), standard software-based drift estimation was not feasible. Consequently, the instrumental drift was calculated manually using a custom **Python-based model**, which is explained in this part.

To accurately identify and quantify this drift trend, continuous time-series measurements were conducted at 30-second intervals at a stable reference station (IPNA basement) during the periods between field survey sessions. By characterizing the specific drift rate for each period, the model allows to isolate and correct instrumental fluctuations from the field survey data, ensuring that the final results reflect true gravitational variations. Continuous files in : / 02_Instrumental_drift / Continuous_files_basement/

The scripts used in this part are :

- /dratf_visualization_drift.ipynb
- /FINAL_TREND_correction.ipynb

Input file of the gravity survey to correct is / Final_Precise_Raw_Gravity_Data.csv

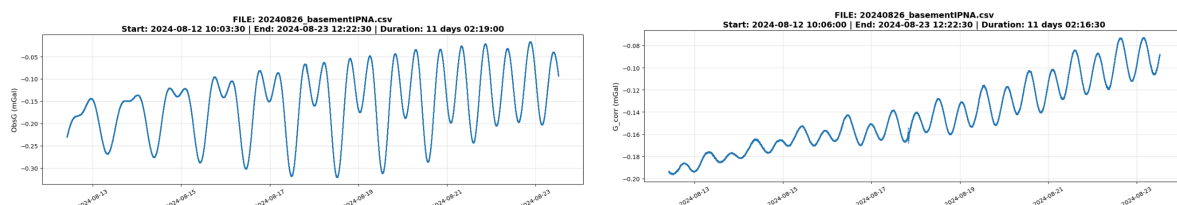
1. First corrections

Apply Solid Earth tides corrections

As raw continuous points are : $\text{ObsG} = \text{feedback correction} + \text{level correction} + \text{temperature correction}$

→ need to apply Earth tide correction (column Earth tide in sheet = Longmann earth tide correction).

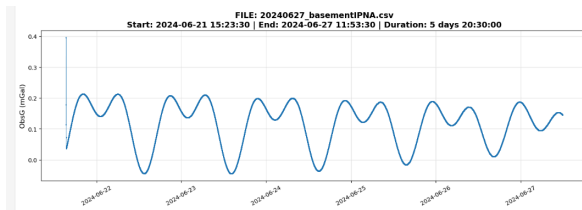
$$\text{In script : } G_{\text{corr}} = \text{ObsG} - \text{Earth tides}$$



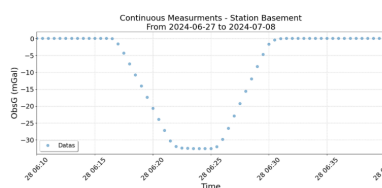
After the correction, there are still some oscillations/wave patterns, but it reduces the oscillations a lot (amplitudes from 0.25 mGal to 0.04 mGal). The remaining oscillation is the sea tide pattern, which is way more complicated to model.

Cleaning the dataset

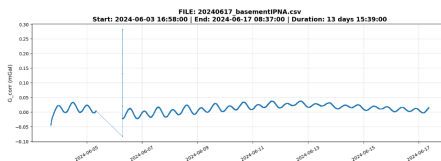
- Delete the 5 first measurements : The first 5 usually are not stable so delete to be more accurate. Gravimeters need a few min to get stable measurements.



- Delete aberrant values (20240708_basementIPNA, on the 28.06 between 6h15 & 6h33 bc went at -30 mgal)

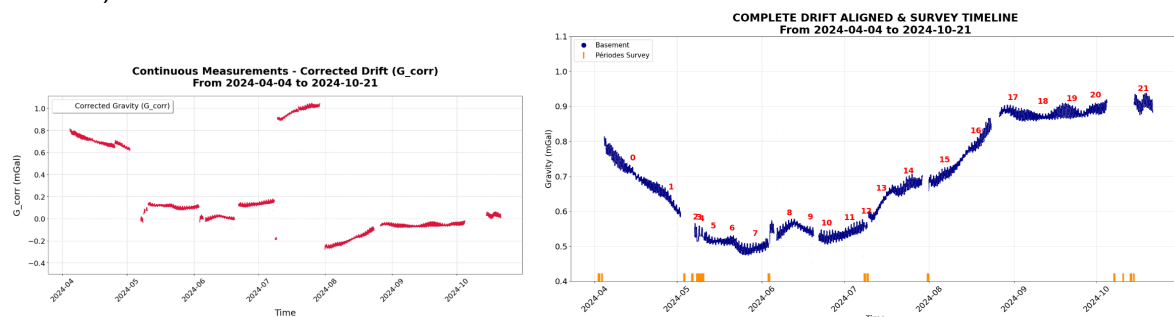


- Gap in the values (20240617_basementIPNA, 23h between 06-05 8h and 06-06 7h). Remove the values before the gap bc not aligned. Sometimes there are jumps after the survey in the first hours.



2. Align the files : offsets

The goal is to align the recording files into a cohesive trend of all the time of the survey (April to October 2024), because there are vertical offsets (gravimeter placed at different heights, tares ??).



→ Developed a "Stitching" script to align the segments:

- Automatic Step:** The script calculates the difference between the end of file N and the start of file N+1 (using a 10-point median for stability).
- Manual Refinement:** We implemented an **Adjustment Dictionary** to manually fine-tune specific offsets where the automatic calculation wasn't

perfect due to noise (zoomed graph to be more precise in the manual adjustment).

3. Calculate drift trend

The complete drift trend is too complicated to be modeled as one polynomial fit. We can break down the drift in multiple periods around the 6 surveys ($\approx 20j$) to model it easier with a second-order polynomial regression. The continuous data files were selected approximately 10 days before and 10 days after the survey.

Resolving the model : TREND fitting method

To approximate the non-linear mechanical drift over time, the model employs a **second-order polynomial regression** ($G(t) = at^2 + bt + c$). This model is resolved using the **numpy.polyfit** function, which implements the **Ordinary Least Squares (OLS)** method to calculate the coefficients (a, b, c) by minimizing the squared differences between the observed gravity data and the predicted temporal trend. The script also evaluates the quality of the fitting by calculating the residuals, as well as R2 and RMSE metrics.

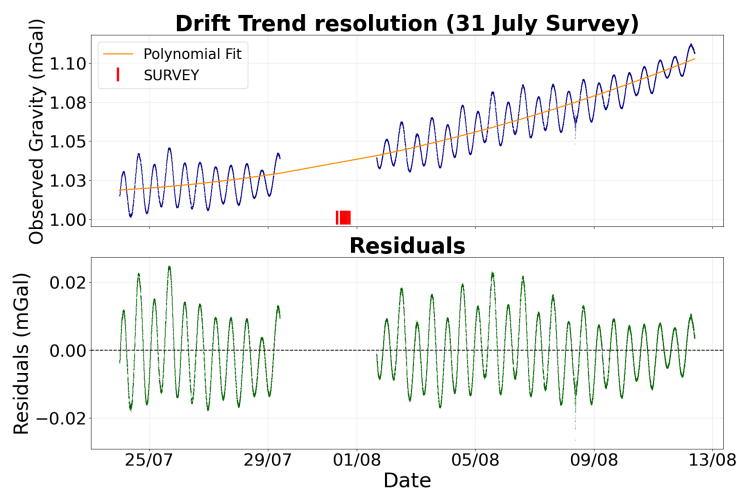


Figure : Plots of the polynomial fit and the residuals.

File summary drift rate and statistics : *in script*

Survey	Start Date	Drift Mean (mGal/j)	RMSE (mGal)	R ²
april	2024-04-02 09:04:51	-0.006367	0.010233	0.934317
may	2024-05-03 14:31:34	-0.005667	0.007943	0.968988

june	2024-06-03 08:30:00	0.003908	0.009332	0.873165
july1	2024-07-08 07:50:27	0.000077	0.009653	0.171948
july2	2024-07-31 08:12:28	0.004333	0.009704	0.877448
oct	2024-10-07 10:02:07	0.000939	0.011365	0.552594

Median : 4.12 µGal/day

Mean : 3.55 µGal/day

4. Drift rate correction

The drift trend formula is then applied to the survey observations data to correct them. The correction is applied from the first measurement of the survey (starting time t0).

We now replaced the raw observed gravity by the drift corrected value (ObsG column).

$$\text{Correction} = \text{ObsG} + \text{calculated drift at time } t$$

Details on the applied correction: 1_ [Survey_details_Data_drift_corrected.csv](#)

Drift corrected values : 2_ [Final_Data_Drift_corrected.csv](#)

5. Height correction : values at ground level

Measurements were conducted over specific benchmarks (reference points at ground level). To ensure proper instrument leveling and stabilization on the field, gravity measurements were recorded at variable heights above the surface (averaging around 25cm). Consequently, the raw observed gravity values reflect the gravity at the sensor's elevation rather than at the actual ground level. To align these observations with ground-level reference values, a height correction was applied using the theoretical Free-Air Gradient of 0.3086 mGal/m. The values were reduced to ground level using the following formula:

$$G_{\text{ground}} = \text{ObsG} + (\text{Height} * \text{Gradient})$$

For an average height of 25 cm, this correction represents an increase of approximately 0.077 mGal (77 µGal) to the observed gravity value.

Finally, the fundamental gravity reductions (including instrumental drift and height correction) have been applied. The **FINAL VALUES** FILE to use for the next corrections is : 3_ [Final_Data_Drift_corrected_Ground_Level.csv](#)

6. Combining Dial and corrected OBSG to prepare for GSADJUST

Output file is / *FINAL_Data_for_GSadjust_BURRIS.csv*